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APOLIPOPROTEIN AI-MILANO: ALTERED LIPID BINDING PROPERTIES IN A HUMAN APOLIPOPROTEIN VARIANT.

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INTRODUCTION

Apolipoproteins AI and AII are the major protein constituents of human high density lipoproteins (HDL). Together they represent approximately 90% of the HDL protein mass. Apolipoprotein AI (apo AI) serves as a cofactor for lecithin: cholesterol acyl transferase (LCAT), the enzyme responsible for the conversion of cholesterol and phosphatidylcholine to cholesteryl esters and lysophosphatidylcholine (Frolich et al., 1982). Apo AI plays an important role in the reverse transport of cholesterol from tissues (Oram et al., 1981) and its plasma level may perhaps be a better predictor of the protection against coronary artery disease, compared to HDL-cholesterol (Avogaro et al., 1979).

Apo AI is a single polypeptide of 243 aminoacids characterized by the absence of cysteine and isoleucine (Brewer et al., 1978); several repetitive sequences, 22 residues long, are present in the terminal part of the apo AI chain (Fitch, 1977). These sequences can assume the structure of an amphipathic α-helix and have been identified as the lipid binding domains in the apo AI molecule (Baker et al., 1975).

Several apolipoproteins can interact in vitro with

different phosphatidylcholines (PC) to form stable lipid-apolipoprotein complexes (Sparrow and Gotto, 1982). Dimyristoylphosphatidylcholine (DMPC) and dipalmitoylphosphatidylcholine (DPPC) have been widely used as tools for investigating the lipid binding properties of apo AI (Jonas, 1984). These PCs associate to apo AI in aqueous solutions, to form complexes of the general size of HDL, readily separable by ultracentrifugal, chromatographic or electrophoretic techiques (Jonas, 1984). The lipidprotein interaction is kinetically controlled and is reversible by heating of by treatment with denaturing agents, such urea and Guanidinium-HCl (Gdn-HCl) (Jonas, 1984).

Polymorphic forms of apo AI occur in the human population, arising from structural variations in the amino-acid sequence of apo AI (Mahley et al., 1984). The first of these variants, the apo AI-Milano (apo AI-M), is characterized by a single cysteine for arginine substitution at residue 173 (Weisgraber et al., 1983). The presence of a cysteine molecule in apo AI-M results in the occurence of AI-M/AI-M dimers and AI-M/AII mixed dimers besides apo AI and AII in the HDL of affected subjects (Weisgraber et al., 1980).

The detection of several variants of human apo AI will help in the understanding the structure-activity relationships in the apolipoprotein molecule. The molecular errors in two of these mutants, i.e. AI-Giessen (Pro-143→Arg) and AI-Marburg (Lys107→0), both unable to activate LCAT in vitro (Mahley et al., 1984), perturb the structure of the amphiphilic region near the substituted residue. Furthermore, the Ala158→Glu variant, in which the 143-162 fragment is more hydrophilic and has a lower helical moment compared to the same region of normal AI, shows altered lipid binding properties (Rosseneu et al., 1986).

The present report focuses on the lipid binding properties of the AI-Milano variant and indicates how the

aminoacid replacement in the mutant modifies the interaction with lipids, altering the secondary structure of the apo AI molecule.

INTERACTION OF APO AI-MILANO WITH PHOSPHOLIPIDS

Purified apo AI and apo AI-M were mixed with DMPC and DPPC at 100/1 PC/AI molar ratio and incubated for 20 hrs at a temperature near the transition T° of the PC; the mixtures were then filtered through a calibrated agarose column. Apolipoprotein and phospholipid coeluted from the column in the form of well defined lipid-protein complexes. DMPC/AI and DMPC/AI-M particles are similar for dimensions (radius: 51.6 vs 51.0 A) and stoichiometries (DMPC/AI molar ratio: 95.4 vs 94.3). By contrast, DPPC/AI-M complexes are slightly larger (radius: 58.2 vs 57.0 A) compared to DPPC/AI particles (Fig. 1).

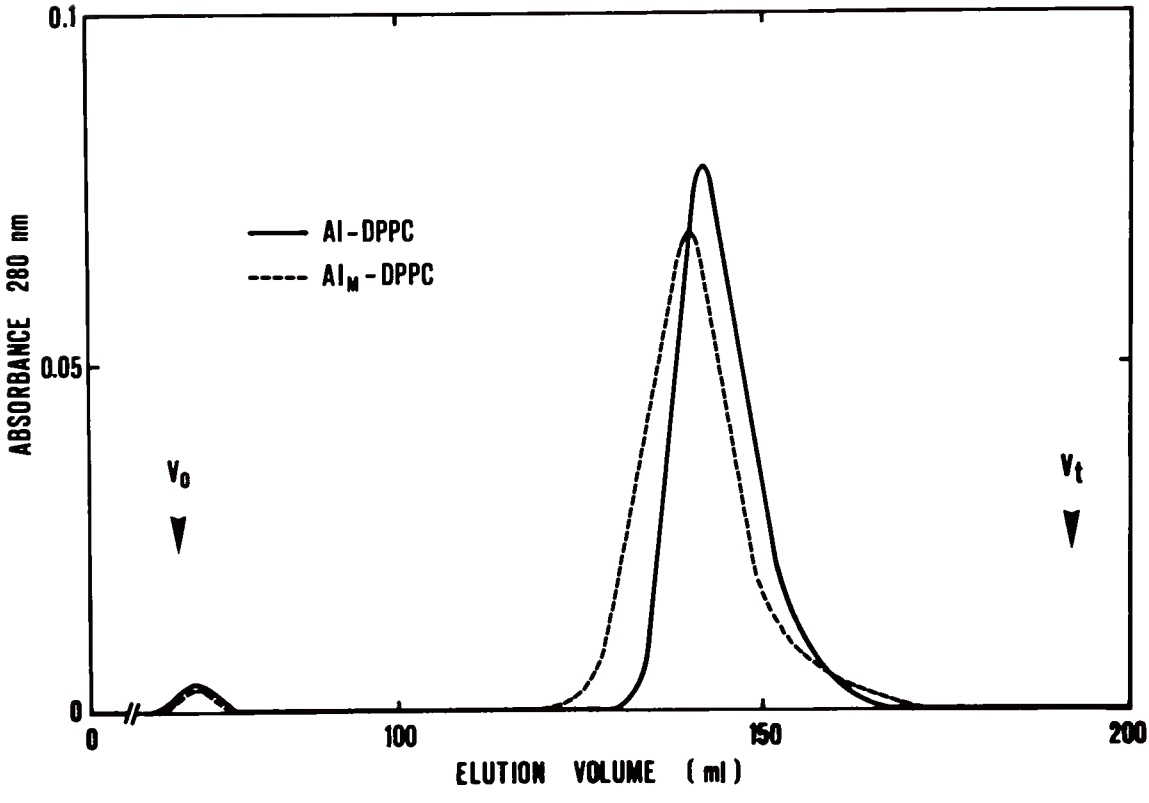


Figure 1. Elution profiles from a 6% agarose column of apo AI/DPPC and apo AI-M/DPPC mixtures.

To investigate the stability of DMPC/AI and DMPC/AIM particles, the purified complexes were exposed to increasing concentrations of Gdn-HCl. Although the same end-product has been obtained after recombination of DMPC with both apolipoproteins, complexes containing apo AI-M denature at lower Gdn-HCl concentrations compared to AI/DMPC particles, suggesting that the mutant apolipoprotein is less tighthly bound to lipids, compared to normal apo AI (Franceschini et al., 1985). On the other hand, a higher affinity of apo AI-M for lipids has been suggested by the 70% increase in the rate of association with DMPC (Franceschini et al., 1985).

PHYSICOCHEMICAL PROPERTIES OF APO AI-MILANO

To evaluate if differences in the secondary structure of the apolipoprotein may be responsible of the altered lipid binding properties of apo AI-Milano, circular dichroism and spectrofluoroscopy analyses were carried out on different preparations of purified apo AI and apo AI-M.

A 2nm red shift in the wavelenght maximum of the fluorescence emission spectrum of apo AI-M was recorded (Franceschini et al., 1985), indicative of an increased exposure to the solvent of the hydrophobic residues in the variant. This experimental finding was confirmed by the prediction of the apolipoprotein conformational features, carried out by the method of Hopp and Woods (1981) (Fig. 2.). The central portion of the apo AI fragment 167-184 turns from a hydrophilic to a hydrophobic character following the aminoacid substitution; the mean hydrophilicity of the segment decreases from 0.89 for normal AI to 0.67 for the mutant.

When the secondary structure of apo AI-M was studied in terms of far-ultraviolet circular dichroism (Franceschini et al., 1985), a lower α-helical content was calculated for the mutant (39%) compared to normal apo AI (54%).

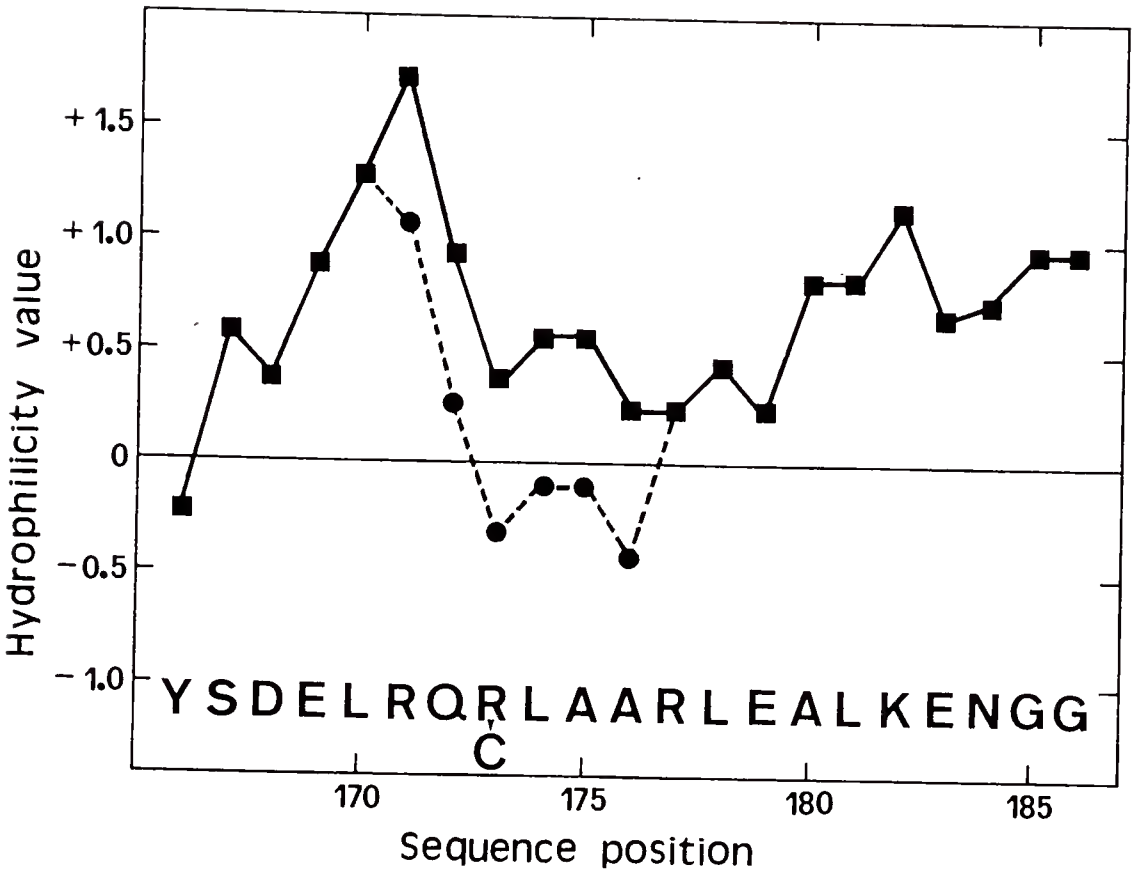


Figure 2. Hydrophilicity analysis of the 167-184 fragment of apo AI (——) and apo AI-M (-----).

Conclusions

In the apolipoprotein AI-Milano, the substitution of Arg173 with cysteine leads to the disappearance of one of the two ion pairs, present in topographically close relationship in the amphipathic fragment between Ser167 and Asn184 (Franceschini et al., 1985), with a resulting loss of amphipathic structure (Bierzynsky et al., 1982). As a consequence, the total secondary structure in the apo AI-Milano will be remodeled, with a decreased total α-helical content and a slightly higher exposure of hydrophobic residues.

The altered secondary structure of the mutant apolipoprotein is probably responsible for the abnormalities in

the interaction of apo AI-M with lipids. Studies carried out by Wetterau and Jonas (1983) showed that dansylated apo AI forms larger complexes with DPPC, compared to those produced by unlabeled apo AI, whereas both apolipoproteins display identical reactivity with DMPC. Furthermore, the chemical modification of amino groups in apo AI resulted in a red shift of the fluorescence wavelenght maximum, in a 10% decrease of α-helix content and in an increase of the overall size of lipid-apo AI complexes, compared to unmodified apo AI (Jonas et al., 1985). These results, comparable with those obtained by us with apo AI-M, suggest that minor modifications of the secondary structure may alter the lipid binding properties of an apolipoprotein. In the case of the apo AI-Milano, these modifications lead to alterations in the association-dissociation process rather than in marked abnormalities in the products of the lipid-apolipoprotein interaction. This is confirmed by the correct binding of purified apo AI-M to HDL after incubation with normal plasma (Brewer et al., 1986). On the other hand the high kinetic flexibility of apo AI-M in the interaction with biological lipid structures may explain the accelerated catabolism of the apo AI-M (Brewer et al., 1986) and, via an increased uptake of tissue lipids, the anti-atherogenic potential of this mutant apolipoprotein (Gualandri et al., 1985).

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